RESEARCH ASSIGNMENT 1

FOUNDATIONS OF DATA ANALYTICS & DATA SCIECE

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SECTION A: DATA FUNDAMENTALS

1 What is a Database? Define in your own words and explain why databases are essential in the modern world. Provide two real world examples of where databases are used.

A Database is a storage place for structured data. As Thomas and Carolyn State, a Database is an organized collection of structured data or information that is electronically saved on a computer system. In the modern era, databases are critical because they support the efficient and reliable administration of extensive datasets. Some of the key reasons why databases are essential in the modern world are

- They impose systematic structure on data, facilitating efficient storage, access, and administration.
- They provide fast retrieval and processing capabilities that underpin rapid decision making and operational continuity.
- They support concurrent multiple user access to shared data while maintaining data integrity.

Two real world examples where databases are used are:

- Studying at a university: There will be a database system that contains information about you as a student, the course you are enrolled in, your financial assistance information, and all your exam results. The university uses database to manage students records
- Bank ATM system: In a bank there is a database that stores account numbers, balances, transaction, and card pins. When you withdraw R500 in sandton, the ATM checks a central database, reduces your balance, logs the transaction. Two minutes later your phone app shows the same new balance because they use the same database.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

A Data Warehouse is a centralized repository that consolidates multiple databases. It is built to store, manage, and analyze large volumes of data gathered from many sources. Unlike regular databases, data warehouses are optimized for querying and analysis

to support business intelligence and decision making (Nordden, 2020).
A company would use a data warehouse for the following reasons:
- Data consolidation: Data warehouse integrates data from multiple sources into a single, unified repository, which enables analysis of the overall business performance.
- Historical insights: A data warehouse saves old data so that businesses can see how things have changed, spot patterns, and make smarter decisions using both past and present information.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

A Data lake is a central repository that can store any amount of data, in any format. It stores structured, semi-structured, and unstructured data in their native form and does not require a schema upfront (Guntupalli, 2023). According to Guntupalli, data lakes and data warehouse differs in their data sources, processing, data quality, and performance, which is explained below;
- Data sources: Both data lakes and data warehouses can store data from unlimited sources. However a data warehouse requires a predefined schema before data can be loaded and accepts only structured data. In contrast a data lake imposes no such requirements and can store any type of data.
- Processing: A data warehouse typically requires processing before storage. Extract, transform, Load (ETL) tools are used to clean, filter, and structure data sets beforehand. In contrast, data lakes hold any data. You have the flexibility to choose if you want to perform processing or not. Organizations use Extract, Load, Transform (ELT) tools. They load the data in the lake first and transform it only when required.
- Data Quality: A data warehouse is generally more reliable because data undergoes processing before loading. Functions such as de-duplication, sorting, and verification can be performed in advance to ensure accuracy. Without such upfront checks a data lake may contain duplicates, errors, or unverified data.
- Performance: A data warehouse is designed for the fastest query performance. In contrast data lake architecture prioritize storage volumes at a lower cost and + you can still access data at reasonable speeds.

A business might prefer a data lake over a data warehouse because it handles diverse data types like structured, semi-structured, and unstructured. For example NETFLIX employs a data lake to house like viewing patterns, user engagements, and search queries. This raw data is later analysed to offer personalized content recommendations and support data informed decisions regarding content creation and procurement.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

As defined by Nordeen (2020) a Data Mart is a subset of the main data warehouse designed to serve a specific line of business such as sales, finance, or marketing. In a data warehouse data is split into a number of simple schemas called data marts, each specialized for a particular analysis activity.
A data mart is closely related to a data warehouse as a smaller focused segment of it. A data mart often extracts and stores a subset of data from the larger data warehouse. A data mart can be dependent on the data warehouse or independent but they usually complement the data warehouse by providing targeted access. Data marts are used by departmental teams, business analysts, and managers and executives.

5. Explain the difference between structured, semi-structured and unstructured Data. Provide one example of each.

Structured data is data that is organized in a consistent format, usually in tables of rows and columns, which facilitates efficient storage, querying, and analysis. Examples of structured data are relational databases and a spreadsheet with defined colums and rows.
Semi-structured data is a data that does not have a rigid structure like structured data, but still contains tags or markers to separate elements, making it somewhat organized and easier to interpete than unstructured data. An example is an email message.
Unstructured data is a data that does not have a predefined format, making it a challenge to process, collect, and analyze. An example here is a video call.

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

According to Simsion et al (2004) Data modeling is the process of creating a structured representation of data and the relationships among data elements within a system. It entails defining how data is stored, organized, and interrelated, typically using diagrams or schemas. Data modeling is important when designing a system that works with data because it improves data quality and consistency by enforcing a standard that should be followed. It also enhance performance as it ensures efficient database design, which directly improves query speed and retrieval. The two types of data models described are conceptual data model and logical data model.

- Conceptual data model: Its a high level model that outlines the overall structure of the data without going into details of implementation. It focuses on what the system contains, defining the main entities and their relationships. An example here is an Entity relationship Diagram (ER) that depicts entities like customers, orders, and products and their relationship.
- Logical Data model: this model provide more detail than the conceptual model, It describes the data in terms of tables, columns, data types, and relationships without considering physical aspects like indexing or storage.

7. What is data mining? How is it used in business? Provide two real world examples of data mining in action.

Data mining is the process of analyzing large datasets to discover patterns, correlations, or trends, Delen (2014). In businesses data mining helps them to make informed decisions, improve customer relationships, optimize operations, and increase profitability by extracting valuable insights from their data. Two real world example of data mining are;

- Netflix uses data mining to identify what customers want to watch, and they can develop new films and series based on those analysis.
- A credit card company like FNB can leverage ~~a~~ it vast warehouse of customers transaction data to identify customers most likely to be interested in a new credit card.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (eg. MySQL, PostgreSQL, Oracle, MongoDB), and explain a suitable use case for each.

Chopra (2010) defined a Database management System as a software that works as an interface between a physical database and the user of the system. With DMS we are able to create, store, retrieve, and manipulate data within a database. Four popular DBMS platforms are;
- MySQL: Its an open source relational DBMS widely used for web applications. It supports structured query language (SQL) and its known for its reliability, ease of use, and strong community support.
- Oracle: It is a powerful, enterprice level relational DBMS designed for large scale applications. It offers advanced features like high availability, scalability, and comprehensive security, making it popular in corporate environments.
- Microsoft SQL: It is a relational DBMS developed by microsoft commonly used in windows environments. It integrates well with other microsoft products and provides tools for business intelligence, analytics, and reporting
- PostSQL: Its an open source, advanced relational DBMS known for its robustness, extensibility, and standards compliance. It supports complex queries and diverse data types, suitable for both small and large applications.

9. What is ETL (Extract, Transform, Load) Explain each step and describe why this process is important in data analytics.

ETL is a standard three step process used to collect data from various sources, clean it, and load it into a central repository, typically a data warehouse, for analysis. Its three steps are explained as;
- Extract: this step involves retrieving data from various sources, which can be databases, files, API's.
- Transform: In this step the extracted data is cleaned, formated, and transformed to fit the targeted system's requirements, this involves removing duplicates, and correcting errors.
- Load: this step involves loading the transformed data into the ~~targeted~~ target database or data warehouse. This step insures that data is properly stored and optimized for querying and analysis.

ETL is important in data analytics because it ensures that data is accurate, consistent, and cleared before analysis which improves the reliability of insights. ETL also consolidate data from multiple sources enabling a comprehensive view for analytics. Regular ETL cycles keep the data warehouse updated with the latest information for real time or near realtime analysis.

10. What are common data storage formats used in analytics (eg CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

- Common separated ~~Notation~~ values (CSV) is a data stored in plain texts separated by commas. It is used for simple, small datasets and ease of use.
- Java Script Object Notation (JSON): Data stored in key-value pairs (nested structures). It is used for semi-structured, nested data and web related data exchange.
- Parquet: is a columnar storage format optimized for the hadoop ecosystem. It is used for large scale analytics, especially with columnar query patterns
- Avro: is a row based binary format that stores the schema within the file. Use avro for compact, schema based sterilization in streaming a data pipeline context.

SECTION B: ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (weak AI) and General AI (Strong AI) Provide one example of each.

According to Dhaman (2026) AI is a field of computer science dedicated to building systems or machines capable of performing tasks that normally require human intelligence. This tasks include learning, reasoning, problem solving, understanding natural language, recognizing patterns, and making decisions. The difference between narrow AI and generic AI are;

Narrow AI (weak AI)

- Developed to handle specific tasks or solve particular problems
- Functions within a restricted domain and cannot apply its abilities outside that scope.

• Lacks true understanding or consciousness, It stimulates ~~the~~ intelligence rather than possessing it.

An example of narrow AI is virtual assistants such as SIRI and ChatGPT.

- General AI (Strong AI)

• A theoretical form of AI that can understand, learn, and apply intelligence accross a wide range of tasks at a human level

• capable of reasoning, solving problems, and adapting new situations without being explicitly programmed for them.

• Does not currently exist and remains the long term objective of AI research.

A example of General AI is Fully autonomous humanoid robot that can learn any skill, like human, without task-specific training.

12. What is machine learning (ML)? How does it differ from traditional programming? Describe the three main types of machine learning: Supervised, Unsupervised, and reinforced learning.

Ravi et al (2023) defined machine learning as a branch of Artificial intelligence that allows computers to learn from data and enhance their performance on tasks without explicit programming for each task. Machine learning algorithms detect patterns in data and leverage those patterns to generate predictions or decisions. In traditional programming the programmer writes explicit rules and instructions to solve a problem, the computer follows these fixed rules to produce an output from given inputs. Machine learning instead of explicit rules, the machine is trained on data. It learns patterns or models from the data which ~~the~~ it then uses to make predictions. Three types of machine learning are:

- Supervised Learning: It is a subfield of machine learning, It involves training machine learning models with the use of data that has been labelled.

- Unsupervised Learning: It is a subfield of machine learning that involves training computers using data that has not been labelled.

- Reinforced Learning: In this subfield of machine learning, a computer can be taught via the process of reinforcement learning how to respond appropriately to a given circumstance in order to maximise the rewards it receives.

13. What is Deep Learning? How does it relate to machine learning and Artificial Intelligence? What are Neural Networks and how do they work

Deep learning is a specialised subset of machine learning that involves training artificial neural networks with many layers to automatically learn and extract complex patterns from large amounts of data. Deep learning is related to both machine learning and Artificial intelligence because it is a specialized subset of machine learning, which is a subset of artificial intelligence. While AI aims to create intelligence machines, and ML uses algorithms to learn from data, deep learning uses multi layered artificial neural networks to automatically extract complex patterns from large unstructured data.
A Neural Network is a machine ~~that~~ learning model that maps inputs to outputs by stacking basic neurons in layers and learning ~~pattern recognition~~ pattern recognizing weights and biases from data.

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real world applications of NLP that you use in everyday life.

Natural Language Processing is a field of artificial intelligence focused on enabling computers to understand, interpret, and generate human language in a way that is meaningful and useful. It combines computational linguistics rule based modelling of human language with machine learning, statistical, and deep learning models to process and analyze large amounts of natural language data. NLP allows machines to perform tasks like language translation, sentiment analysis, speech recognition, and text summarization by understanding the structure and meaning of language, Deri et al (2023). Three real world applications of NLP that many people use in everyday life are;
- Virtual assistants: Tools like Siri and Google assistant use NLP to understand spoken commands, interpret user intent and respond naturally.
- Email spam filtering: Email services apply NLP techniques to analyze the content of emails and distinguish between legitimate messages and spam or phishing attempts.

- Language translation; services like google translate use NLP to automatically translate text or speech from one language to another, enabling communication across language barriers.

15. What is a Large Language Model (LLM) How is it different from a standard machine learning model? Name two well known LLMs and explain what they are used for.

Large language model is a type of AI built on deep learning and transformer architecture that understands and generates human like text by analyzing vast datasets. Two well known LLMs:
- Generative Pre-trained transformer (GPT): they are designed to understand and generate human like text based on the input they receive. An example of GPT is a customer support chatbots to answer queries
- Bidirectional Encoder Representations from transformer: It is primarily used for understanding the context of words in a sentence by looking at both the left and right context simultaneously.

16 What are generative AI? How does it work at a high level Name three tools that use Generative AI and describe what each one does.

Dhamani et al (2026) described Generative AI as a category of AI systems use machine learning models to create original content such as text, image, videos, audio, or software code in response to a users prompt or request. At a high level Generative AI works by analyzing large amounts of existing data to learn underlying patterns, structures and relationships.
Dua (2004) explained three tools that uses Generative AI as;
- ChatGPT: It generates human like text for conversations, writing assistance, content creation, etc. It can answer questions, write essays, and simulate dialog based on a user input.
- Github Copilot: It assists software developers by generating code snippets and suggestions based on comments or partial code.
- DALL·E: It creates original images from textual descriptions. Users provide a prompt and DALL·E generates an image matching that description.

17. What is training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

Training dataset is a collection of data used to teach AI model how to perform a specific task. This dataset contains examples with inputs, and often corresponding outputs that AI uses to learn patterns, relationships, and make predictions or decisions. Quality of training data is important when building an AI model because it directly dictates the performance, accuracy, and reliability of an AI model. High quality data is accurate and consistent. If the training data is biased or incorrect, users may lose confidence in the AI system, it may lead to inaccurate results, and also poor generalization.

18. What is computer vision? How do machines 'see' and interprete images? Provide two real world ~~examples~~ use cases of computer vision.

According to Szeliski (2010) a computer vision is a field of artificial intelligence that enables computers to interpret and understand the visual world, analyze images and vides to identify objects, scenes, and patterns. Machine 'see' by converting visual data into digital pixels and using machine learning models to detect features and recognize patterns. Two real world use cases of a computer vision are;

- Healthcare: medical image analysis, computer vision algorithms are used to analyze x-rays, CT scans, and MRIs to detect abnormalities like tumors, fractures, or diseases faster and with high accuracy.
- Quality control in manufacturing: Cameras on production lines inspect products in real time to identify anomalies, improving efficiency and reducing human error.

SECTION C: API'S, TOOLS & DEVELOPER CONCEPTS

19. What is an API (Application Programming Interface)? Explain what it is using a simple real world analog, why are APIs important in modern software and data systems?

API in simple terms is a way for different systems or applications to communicate with each other. For example in a restaurant for you to get food three parts must work together which are:
- The customer (you): you give your order to the waiter
- The kitchen (system): is the place that prepares or cooks the food
- The waiter (API): is the messanger, she delivers the message.

Importance of API's
- APIs enables different systems, services, or applications to work together seamlessly
- They ensure data looks the same on every device
- They allow developers to reuse existing functionality rather than building from scratch.

20. What is an API key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

API key is a unique identifier used to authenticate application and systems attempting to access other software or systems via an application programming interface.

API key purposes
- Authentication: It verifies the identity of the requester
- Authorization: It controls access to API resources
- Usage tracking: It monitors how and how much the API is used.
- Security: It helps prevent unauthorized access of the API

API key should be stored and protected by the following:
- Keeping it secret, never expose API keys in public code repositories
- Use environment variables to store key securely
- Set restrictions so the key only works from specific IP addresses.
- Rotate key regularly to minimize damage is one is stolen.

Risks of public exposure
- Unauthorized access
- Data theft and manipulation
- Service disruption.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, DELETE) used in REST API? Briefly describe each method

REST stands for Representational State Transfer. A REST API is an interface that two computer systems use to exchange information securely over the Internet. It uses HTTP requests to perform operations on resources which are identified by URLs. Main HTTP methods are described as follows;
- GET - It retrieves data from the server
- POST - It sends data to the server to create a new resource.
- PUT - It updates an existing resource or create it as if it does not exist.
- Delete - It removes a specified resource from the server.

22. What is JSON (Javascript Object Notation) why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON Object.

Jason is a text based format for storing and exchanging data in a way that is easy for humans to read and write, and easy for machines to parse and generate.
Why is JSON commonly used in API's and data exchange?
It is simple and light weight, making data transfer efficient
- It is easy to parse and generate in most programming languages
- It supports complex ~~databases~~ data structures like objects & arrays.
- It is human readable, which helps with debugging and development

Example:
```
{
  "name": "Boitshelo",
  "age": 37,
  "is student": True
  "course": ["Data analytics"]
}
```
The object describes a person named Boitshelo, who is 37 years, is a student, and is doing data analytics course.

23 What is prompt engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of poor written prompt and an improved version of each.

Prompt engineering is the process of creating and refining the instructions given to an AI model to improve the accuracy and relevance of its responses. It is important when working with AI tools because it helps unlock the full potential of AI tools for various tasks. It saves time and effort by minimizing the need for repeated corrections.

Examples of poorly written and improved versions:

- Poor prompt: "tell me about ~~yourself~~" history".
  improved: "Give me a brief summary of the causes and side effects of world war II."
- Poor prompt: "write a story".
  improved "write a short story about a young girl who discovers a secret garden in her backyard".

24. What is a MAP? Explain what it does in programming and data processing. Provide a simple example in either python or JavaScript.

A MAP function is a tool used in programming and data processing to transfer every item in a list or collection by applying the same rule to each other, and return a new collection containing the results.

What it does:

- It takes function and collection as inputs
- applies the function to each element of the collection
- It ~~provide~~ produces a new collection with the transformed elements
- It does not modify the original collection

Example in python

```
Numbers = [1, 2, 3, 4]
squared = list (map (lambda x: x**2, numbers))
print (squared) # output : [1, 4, 9, 16]
```

25. What is cloud computing? Name the three main cloud services models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model

Cloud computing is the delivery of computing services including servers, storage, databases, networking, and analytics over the internet.
Three main cloud models
- IaaS (Infrastructure as a Service): It provides virtualized hardware, such as computing resources, storage, and networks. You manage the operating system and applications while the provider takes care of the physical hardware and virtualized infrastructure
- PaaS (Platform as a Service): Abstracts much of the infrastructure management, providing a platform where you can build and deploy applications without worrying about underlying systems
- SaaS (Software as a Service): It provides fully managed software applications over the internet. You interact with the software directly with no need to manage any infrastructure or platform.

26 What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are?

A version control system is a system that records any changes made by the software developers, Zolkifli (2018). Git is the most popular distributed version control system. Unlike older systems, every user has a full copy of the project history on their local machine. It is widely used by data analysts and developers because,
- It efficiently manages projects of any size
- It supports branching and merging, enabling parallel development
- It works offline and syncs changes when connected
- It integrates well with platforms like Github, Gitlab and Bitbucket for collaboration and code sharing.

Commit is a snapshot of changes made to the files in the project at a particular point in time.
Branch is a separate line of development that diverges from the main codebase.

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among data analysts and data scientists?

Jupyter is an open source web application that allows users to create and share documents containing live code, equations, visualizations, and narrative text.
Why is it popular among data analysts and data scientists
- It allows step by execution and immediate feedback
- easily integrates with libraries
- they can be shared and version controlled

28. What is Python? Why is it considered one of the most popular programming languages for data analytics and AI? Name four python libraries commonly used in a data work and briefly describe each.

Python is a programming language widely used by data scientists. Its a high level, interpreted programming language known for its readability or versatility.
Python is considered one of the most popular language because of:
- Easy to learn and use
- rich ecosystem
- Strong community support
- Integrates easily with other languages
The four commonly used python libraries in data work are:
- Pandas: provides optimized and flexible data structures that you can use to manipulate time series data and structured data.
- NumPy: is a popular library that developers use easily to create and manage arrays, manipulate logical shapes, and perform linear algebra operations.
- Matplotlib: this is a comprehensive library for creating static, animated, and interactive visualizations.
- Scikit-learn: A machine learning library that includes simple and efficient tools for data mining and analysis.

D. NumPy - Is a popular library that developers use to easily create and manage arrays, manipulate logical shapes, and perform linear algebra operations. NumPy supports integration with many languages such as C and C++.

). Matplotlib - This is a comprehensive library for creating static, animated, and interactive visualizations. It allows you to turn dry numbers into line graphs, bar charts, and histograms to help communicate insights.

). Scikit-learn - A machine learning library that includes simple and efficient tools for data mining and analysis, such as classification, regression, and clustering algorithms.

SECTION D: DATA ~~ETHICS~~, GOVERNANCE & CAREERS

1). What is data privacy? Why is it important? Research e GDPR (General Data Protection ~~Act~~ Regulation and explain what it is and which countries it applies to.

Data Privacy is the protection of personal data from unauthorized access and misuse, allowing individuals control over their information. Its importance → It is essential for safeguarding rights, preventing identity theft, and fostering trust between organizations and customers.

General Data Protection Regulation (GDPR) - Is a key ~~law~~ EU law effective since May 2018, aimed at protecting the privacy of EU citizens by regulating data handling practices. It requires clear consent from individuals and provides rights to access, correct, and delete personal data. The GDPR applies to all EU member states and organizations outside the EU that interact with EU citizens, setting a global benchmark for data privacy standards. GDPR.E (2018)

3). What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

Data Ethics ~~refers to~~ e moral principles and guidelines that govern e collection, analysis, sharing, and use of data, especially when it involves personal or sensitive information. It focuses on ensuring fairness, privacy, transparency, accountability, and respect for individual's rights in a data related practices.

3 examples of unethical use of data or AI

1) Biased AI algorithms: e.g. an AI hiring tool that systematically favours certain demographic groups (e.g. based on gender or ethnicity)

Problem: This leads to unfair discrimination, perpetuates social inequalities

and or to gain opportunities

2. Unauthorized data collection & surveillance - eg collecting personal data from users without their explicit consent or knowledge.

Problem: violates privacy rights and trust, potentially exposing sensitive info to misuse or breaches.

3. Deepfake technology for misinformation - eg creating realistic ~~fake~~ but fake videos or audio to manipulate public opinion or defame individuals

Problem: undermines trust in information, spreads falsehoods, and can cause significant harm to individuals or society.

3) What is data governance? Define it and explain 3 key components of a good data governance framework

Data governance is a set of policies, processes, standards and controls that ensure the effective and responsible management of an organization's data assets. It aims to maintain data quality, security, privacy, and compliance while enabling data to be used as a valuable resource for decision-making.

Key components of a good data governance framework

1. Data quality management - ensures data accuracy, completeness, and reliability through validation and cleaning processes.
2. Data stewardship - assigns accountability by designating individuals or teams responsible for managing and protecting specific data assets
3. Data policies & standards - Defines rules and guidelines for data usage, access, privacy, security, and compliance with legal & regulatory requirements.
4. Data security & privacy - implements measures to protect data from unauthorized access, breaches, and misuse, while respecting privacy rights.
5. Data lifecycle management - manages data from creation to deletion, including storage, archiving, and retention policies
6. Compliance and Risk management - Monitors adherence to regulatory requirements and ~~assessment~~ assesses risks related to data management
7. Data architecture & technology - Establishes a technical infrastructure and tools that support data governance activities

4) What is Bias AI? Explain what it means for an AI model to be biased. Provide a real world example of AI bias & describe potential consequences.

Bias AI refers to a systematic and unfair favoritism or discrimination that occurs when an artificial intelligence system produces outputs that

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